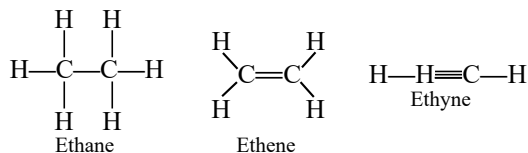


Organic Chemistry: Some Basic Principles and Techniques

SHORT NOTES

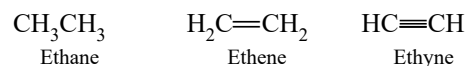
Complete, Condensed and Bond-line Structural Formulas

- Ethane (C_2H_6), ethene (C_2H_4), ethyne (C_2H_2) can be represented by the following structural formulas. Such structural representations are called complete structural formulas.



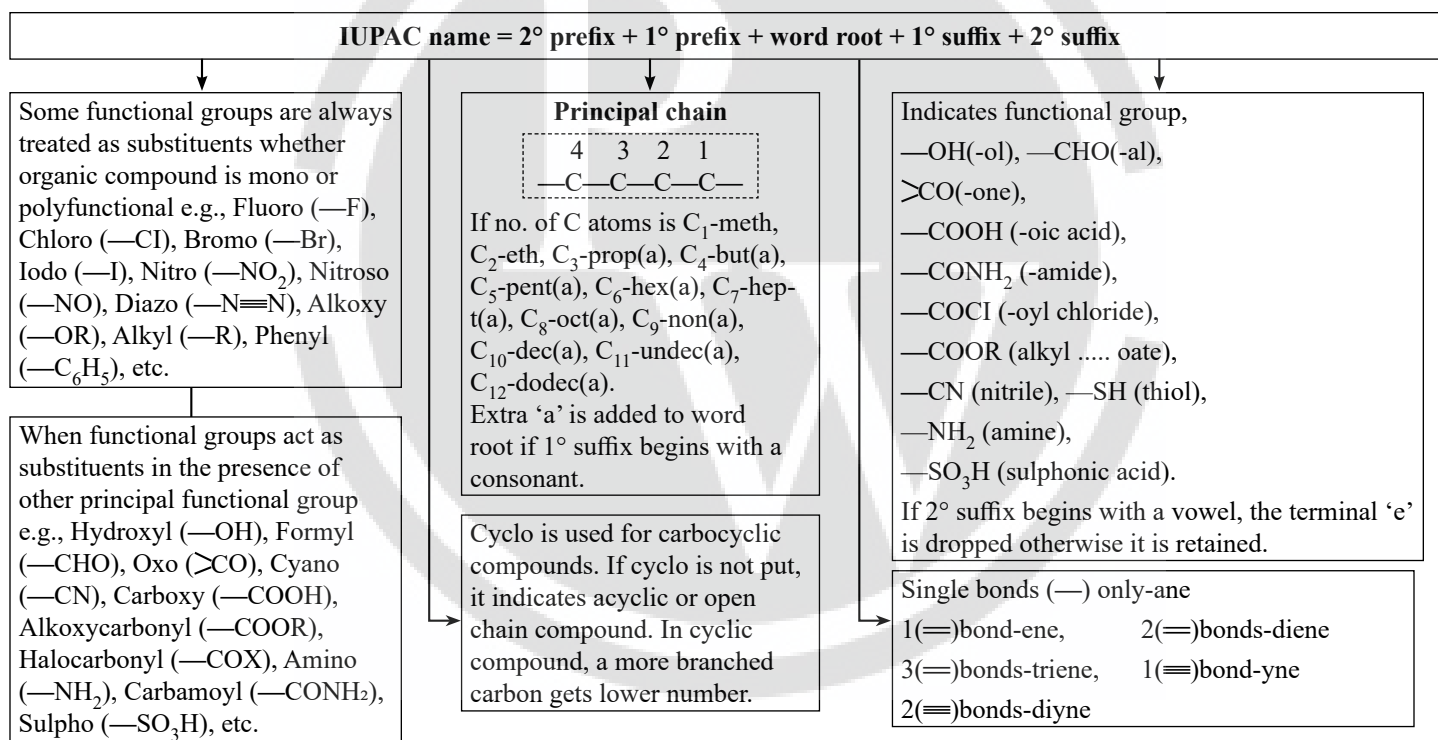
- In condensed structural formulae the bonds to each carbon are omitted, but each distinct structural unit (group) is written

with subscript numbers designating multiple substituents, including the hydrogens.

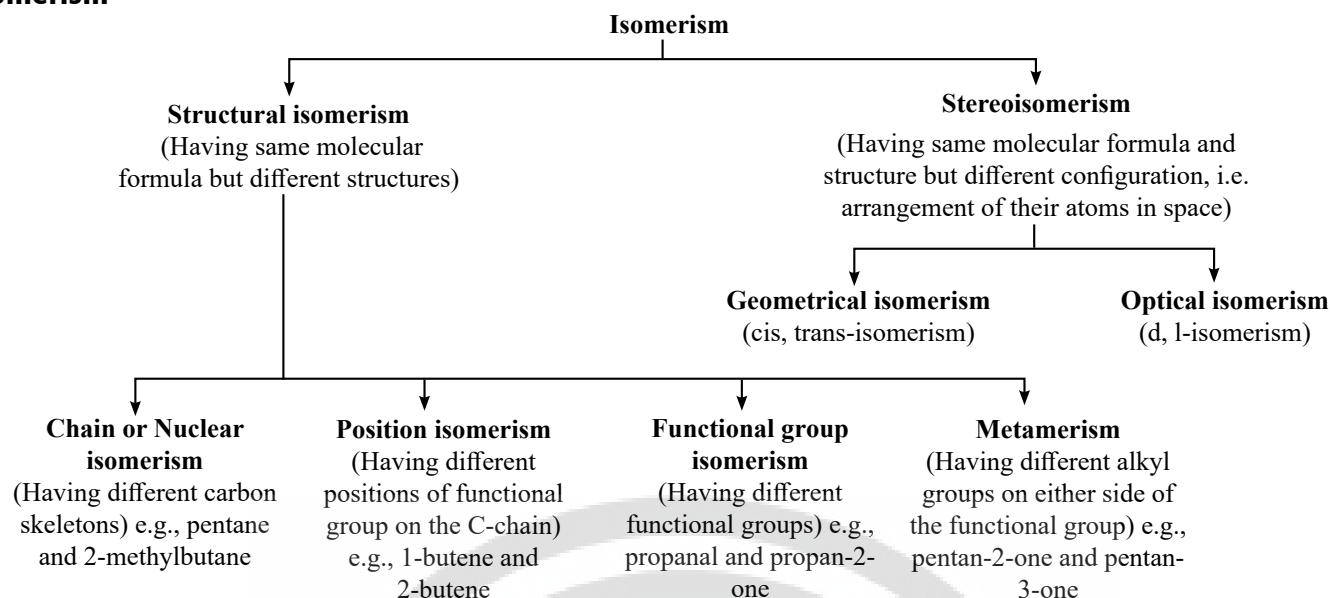


Nomenclature of organic compounds

- The complete IUPAC name of an organic compound consists of five parts. Secondary prefix + Primary prefix + Word root + Primary suffix + Secondary suffix
- Preference order (decreasing priority) for selection of a principal functional group in polyfunctional compounds is $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{COOR}$, $-\text{COCl}$, $-\text{CONH}_2$, $-\text{CN}$, $-\text{HC}-$, $-\text{O}$, $>\text{C}=\text{O}$, $-\text{OH}$, $-\text{NH}_2$, $>\text{C}=\text{C}<$, $-\text{C}\equiv\text{C}-$.



Isomerism



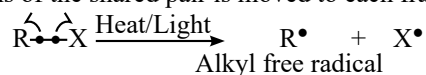
Configurational nomenclature in geometrical isomerism		
Configuration	Criteria	Remarks
cis/trans	Similarity of groups	If the two similar groups are on same side of restricted bond the configuration is cis otherwise trans.
E/Z	Seniority of groups	If the two senior groups are on same side of restricted bond the configuration is Z (Z = zusammen = together) otherwise E (E = entgegen = opposite).

Table: Enantiomers vs Diastereomers

	Properties	Enantiomers	Diastereomers
(1)	Molecular formula	Same	Same
(2)	Structural formula	Same	Same
(3)	Stereochemical formula (structure formula with orientation)	Different	Different
(4)	Dipole moment	Same	Different
(5)	Physical properties (m.p., b.p., density, solubility, refractive index etc.)	Same	Different
(6)	Specific rotation	Different sign but same magnitude	Different
(7)	Chemical properties		
	(1) with optically inactive compound	Same	Different
	(2) with optically active compound	Different	Different

Fission of a Covalent Bond

- ❖ **Homolytic fission (symmetrical):** Free radicals are formed by homolytic cleavage of covalent bond in which one of the electrons of the shared pair is moved to each fragment.



- ❖ **Heterolytic fission (unsymmetrical):** Positive and negative ions are formed by heterolytic cleavage of covalent bond in

which both the electrons of the bonding pair are moved to one of the fragments (i.e., towards more electronegative atom).

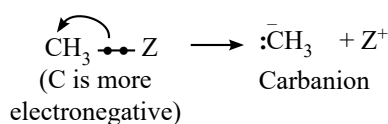
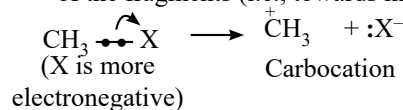


Table: General Properties of Reaction Intermediate

Property	Carbocation	Carbon Free radical	Carbanion
Formation	$C-X \rightarrow C^+ + X^-$	$C-X \rightarrow C^\bullet + X^\bullet$	$C-X \rightarrow C^- + X^+$
Formed by	Heterolytic fission	Homolytic fission	Heterolytic fission
No. of e^- in valence shell	6	7	8
Octet	Incomplete	Incomplete	complete
e^- rich/deficient	Deficient	Deficient	rich
Stabilized by	e^- donating groups	e^- donating groups	e^- withdrawing groups
Unpaired e^-	0	1	0
Represented by	$\begin{array}{c} \\ -C^{\oplus} \\ \end{array}$	$\begin{array}{c} \\ -C^{\bullet} \\ \end{array}$	$\begin{array}{c} \\ -C^{\ominus} \\ \end{array}$
hybridisation	sp^2	sp^2	sp^3
shape	Trigonal planar	Trigonal planar	Pyramidal

Electronic Displacement in Organic Compound

❖ **Inductive effect:** Displacement of σ electrons along a saturated carbon chain when an electron donating (+I effect) or electron withdrawing (-I effect) group is attached at the end of the carbon chain.

+ **Decreasing order of -I effect:** $-\text{NO}_2 > -\text{CN} > -\text{COOH} > -\text{F} > -\text{Cl} > -\text{Br} > -\text{I} > -\text{H}$

+ **Decreasing order of +I effect:** $(\text{CH}_3)_3\text{C}- > (\text{CH}_3)_2\text{CH}- > \text{CH}_3\text{CH}_2- > \text{CH}_3- > \text{D}- > \text{H}-$

❖ **Electromeric effect:** It is a temporary effect that involves complete transfer of π electrons of a multiple bond towards one of the bonded atoms in the presence of an attacking reagent.

+ +E effect when electron transfer takes place towards the atom to which the attacking reagent gets attached.

+ -E effect when electron transfer takes place away from the atom to which the attacking reagent gets attached.

❖ **Resonance or Mesomeric effect:** It is defined as the polarity produced in the molecule by the interaction of two π bonds or between a π bond and a lone pair of electrons present on adjacent atom. This effect is transmitted through the chain. If the chain or cyclic system consist of alternate single and double bonds then it is termed as a conjugated system.

+ +R or +M effect: Groups which show this effect are $-\text{OH}$, $-\text{OR}$, $-\text{OCOR}$, $-\text{SH}$, $-\text{SR}$, $-\text{NH}_2$, $-\text{NHR}$, $-\text{NR}_2$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, etc.

+ -R or -M effect: Groups which show this effect are $-\text{COOH}$, $\text{C}=\text{O}$, $-\text{CHO}$, $-\text{COOR}$, $-\text{CN}$, $-\text{NO}_2$, etc.

❖ **Hyperconjugation:** It involves delocalisation of σ electrons of C-H bond of an alkyl group directly attached to an atom of unsaturated system or to an atom with an unshared p-orbital. Hyperconjugation is a permanent effect. For an alkyl group attached to unsaturated system it decreases with decrease in number of α -hydrogens:



Aromaticity

